Comparing Compact and Deep Convolutional Networks for CIFAR-100 Classification

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***Abstract*—We train convolutional neural networks from scratch on CIFAR-100 and compare a compact CNN with a deeper, wider CNN under three joint learning-rate, dropout, and L2 settings. Data are split into 45,000 training, 5,000 validation, and 10,000 test images. Training uses Adam, augmentation, batch normalization, dropout, L2 regularization, learning-rate reduction, and validation-loss early stopping (patience 8). The validation-selected Deep C model reaches 63.04% top-1 test accuracy (95% Wilson CI 62.09–63.98%), 36.96% generalization error (CI 36.02–37.91%), and 88.66% top-5 accuracy, 26.13 points above the best Compact model. Confusion matrices show stronger class-wise discrimination for the deep network, and Integrated Gradients (IG) shows that a confident fine-grained error can still be driven by object pixels. Capacity matters most; less regularization helps Deep only slightly, while a lower learning rate with heavier regularization hurts both architectures.**

***Index Terms*—CIFAR-100, convolutional neural networks, regularization, confidence intervals, confusion matrices, Integrated Gradients**

I. Introduction

CIFAR-100 contains 100 fine-grained classes of low-resolution 32×32 RGB images [1], with many classes sharing color, texture, and background. This makes it a useful testbed for how model capacity, optimization, and regularization affect generalization, and top-5 accuracy is informative alongside top-1 because many errors are between visually similar classes.

We train a Compact CNN and a Deep CNN from scratch (no pretrained weights) under joint learning-rate, dropout, and L2 settings, and we use each result to guide the next experiment: Setting B underfit, so Setting C was added with lighter regularization. Models are compared by validation loss, then evaluated once on 10,000 held-out test images with top-1/top-5 accuracy, 95% Wilson confidence intervals, training curves, normalized confusion matrices, and Integrated Gradients (IG) [2]. The validation-selected Deep C model achieves 63.04% top-1 (CI 62.09–63.98%) and 88.66% top-5 accuracy, 26.13 points above the best Compact model. We conclude that capacity dominates performance, lighter regularization adds little, and heavier regularization with a lower learning rate hurts both architectures.

II. Problem Formulation and Data

Each image is $x \in [0,1]^{32\times32\times3}$ and each label is a one-hot vector $y \in \{0,1\}^{100}$. The network outputs a softmax vector $p_\theta(x)$ and predicts $\hat{y} = \arg\max_k p_{\theta,k}(x)$. Training minimizes categorical cross-entropy plus an L2 penalty on convolutional and hidden dense kernels,

$$\mathcal{L}(\theta) = -\frac{1}{B}\sum_{i=1}^{B}\sum_{k=1}^{100} y_{ik}\log p_{\theta,k}(x_i) + \lambda \sum_{W\in\mathcal{R}} \|W\|_F^2, \quad (1)$$

where B is the batch size and $\mathcal{R}$ the regularized weight matrices. The final softmax layer is not penalized, so reported Keras losses include the L2 terms.

CIFAR-100 is loaded through TensorFlow Datasets (50,000 training and 10,000 test images). The training set is split with a deterministic, stratified 90/10 split (seed 42) into 45,000 training and 5,000 validation images, giving 450/50/100 training/validation/test images per class. Pixels are scaled by 1/255. Augmentation is applied to training images only: padding to 40×40, a random 32×32 crop, and a random horizontal flip.

TABLE I
Compact and Deep CNN Architectures

Stage	Compact	Deep
1	C_{32}, C_{32}, P, D(d/2)	C_{32}, C_{32}, P, D(d/2)
2	C_{64}, P, D(d)	C_{64}, C_{64}, P, D(d/2)
3	—	C_{128}, C_{128}, P, D(d/2)
Aggregation	GAP (64)	GAP (128)
Hidden	Dense 128, ReLU, D(d)	Dense 256, ReLU, D(d)
Classifier	Dense 100, softmax	Dense 100, softmax
Parameters (total)	50,244	347,076

III. Architectures and Training Protocol

Both networks use 3×3 stride-1 same-padded convolutions without bias, each followed by batch normalization (BN) and ReLU, and 2×2 max pooling. Small kernels compose local features efficiently, BN stabilizes and speeds optimization (so the bias is redundant), ReLU avoids saturating gradients, pooling reduces spatial cost, and GAP limits dense-layer parameters and overfitting. Table I gives the structures, where C_f is a convolution (f filters) with BN and ReLU, P is max pooling, D is dropout, and GAP is global average pooling. Deep has about 6.9 times more parameters than Compact (347,076 vs. 50,244) and differs in depth, width, dense size, and dropout placement, so comparisons support the higher-capacity design as a whole rather than depth alone.

All models use Adam, chosen for its robustness to learning-rate scale on deep convolutional networks, with batch size 128. Regularization combines L2 weight decay, dropout, and augmentation (crop and flip preserve the class label), in

TABLE II
HYPERPARAMETER SETTINGS (LEARNING RATE, DROPOUT d, L2 λ)

Setting	LR	d	λ	Compact	Deep
A	1×10^{-3}	0.30	1×10^{-4}	yes	yes
B	3×10^{-4}	0.50	5×10^{-4}	yes	yes
C	1×10^{-3}	0.20	5×10^{-5}	no	yes

addition to early stopping; the 90/10 stratified split keeps all classes balanced in validation. Table II lists the joint settings. A and B were run for both architectures. B pairs a lower learning rate with heavier regularization and underfit, so Setting C, which uses *lighter* regularization than A, was added for Deep only. A uses a standard Adam rate with moderate dropout and L2 as a baseline; B tests whether stronger constraints help; and the widths double per stage (32, 64, 128), while Deep's dense layer is twice the size of Compact's (256 vs. 128), following common CNN practice, to give Compact and Deep clearly different capacities.

Training runs for at most 150 epochs. Early stopping monitors validation loss (patience 8, minimum improvement 10^{-4}, best weights restored). All five runs stopped through early stopping before the epoch limit, and checkpoints and model selection use validation loss only; the test set is never used for selection. Because an epoch counts as an improvement only if validation loss falls by at least 10^{-4}, a later epoch with a smaller gain does not reset the patience counter, so Compact A and Deep B stopped fewer than 8 epochs after their lowest validation loss (Table III).

IV. EXPERIMENTAL RESULTS

A. Metrics and confidence intervals

Top-1 accuracy requires the highest-probability class to be correct; top-5 accuracy requires the true class among the five most probable; generalization error is $1 -$ accuracy. For $\hat{p}$ correct predictions out of $n = 10{,}000$ test images, the 95% Wilson interval is

$$\frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \tag{2}$$

with $z = 1.96$; applying it to incorrect predictions gives the error interval. These intervals capture finite test-sample uncertainty for a fitted model, not variation across training seeds.

B. Best model and architecture effect

We select the best model by minimum validation loss because it uses data never trained on, is not tied to a threshold, and keeps the test set unbiased. Deep C has the lowest validation loss (1.4728) and the best test performance (Table III): 6,304 of 10,000 images correct, 63.04% top-1, 36.96% error, and 88.66% top-5. Its top-5 accuracy is 25.62 points above its top-1 accuracy, so many misses still rank the true class among the top five, which matters in a 100-class fine-grained problem.

Capacity has the largest effect. Deep exceeds Compact by 25.84 points under A (62.75% vs. 36.91%) and 27.21 points under B (55.01% vs. 27.80%), and the best Deep model exceeds the best Compact model (Compact A) by 26.13 points; these gaps are far larger than the interval widths of about 2 points.

C. Effect of hyperparameters and regularization

Setting A beats B for both architectures: Compact falls from 36.91% to 27.80% (9.11 points) and Deep from 62.75% to 55.01% (7.74 points). Fig. 1 shows B with higher loss and lower accuracy in both training and validation, consistent with underfitting from a smaller learning rate plus heavier dropout and L2. Because the three factors change together, these results establish the effect of each *joint configuration*, not of any single hyperparameter.

Lightening regularization (A to C) improves Deep only slightly: validation loss drops from 1.5199 to 1.4728 and test accuracy rises 0.29 points (62.75% to 63.04%). That difference is much smaller than the CI width and the two intervals overlap almost completely, so we treat A and C as comparable in accuracy, with C preferred by validation loss. C also shows the largest train–validation gap (about 68.4% vs. 63.1% accuracy at its best epoch, with augmentation and dropout still active in training). By the definition, overfitting means a model that is better on training data but worse on unseen data than another model. C is better than A on training data but not worse on validation or test data, so this gap does not show harmful overfitting; it only shows that C is the least constrained.

Validation accuracy exceeds training accuracy for Compact A and Deep B because dropout and augmentation are active only in training; Compact B shows low accuracy with almost no gap, more consistent with limited capacity and heavy constraints than overfitting. Deep A and C show modest gaps without validation degradation, and since the underfit runs (both B settings and Compact A) have low training accuracy, their weakness is insufficient fit, not overfitting.

D. Normalized confusion matrices

For each architecture we analyze the validation-selected model: Compact A and Deep C. The matrix is row-normalized, $\widetilde{C}_{ij} = C_{ij} / \sum_j C_{ij}$, so the diagonal is per-class recall (each test class has 100 images, so one correct image adds 0.01).

Deep C shows a stronger, more continuous diagonal than Compact A, so its gain is spread across many classes rather than a few. Both matrices keep off-diagonal mass among visually similar low-resolution classes, so fine-grained errors remain.

V. INTEGRATED GRADIENTS ANALYSIS

For input x, baseline x', and target output F_c, IG attributes feature j as [2]

$$\mathrm{IG}_j(x) = (x_j - x'_j) \int_0^1 \frac{\partial F_c(x' + \alpha(x - x'))}{\partial x_j}\, d\alpha. \tag{3}$$

We use a black baseline, the softmax probability of the predicted class as F_c, and 64 trapezoidal steps. Absolute

TABLE III
FINAL HELD-OUT CIFAR-100 RESULTS. INTERVALS ARE 95% WILSON INTERVALS. TEST LOSS INCLUDES L2 PENALTIES. THE BEST MODEL BY VALIDATION LOSS IS IN BOLD.

Model	Epochs / Best	Min. Val. Loss	Test Loss	Top-1 (%)	Top-1 CI (%)	Error CI (%)	Top-5 (%)
Compact A	69 / 66	2.4864	2.4600	36.91	[35.97, 37.86]	[62.14, 64.03]	69.44
Compact B	91 / 83	2.8950	2.8913	27.80	[26.93, 28.69]	[71.31, 73.07]	61.17
Deep A	105 / 97	1.5199	1.5007	62.75	[61.80, 63.69]	[36.31, 38.20]	88.43
Deep B	112 / 107	1.8190	1.7872	55.01	[54.03, 55.98]	[44.02, 45.97]	84.21
Deep C	**97 / 89**	**1.4728**	**1.4601**	**63.04**	**[62.09, 63.98]**	**[36.02, 37.91]**	**88.66**

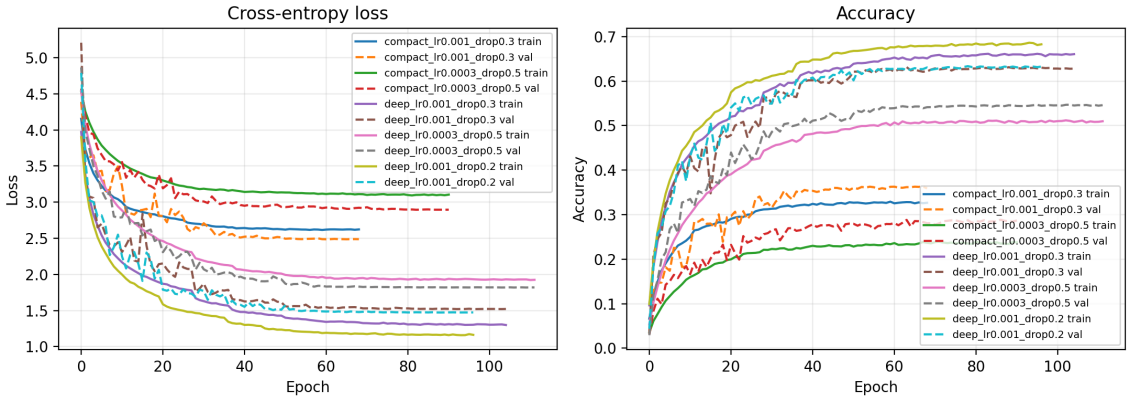


Fig. 1. Training (solid) and validation (dashed) cross-entropy loss including L2 (left) and accuracy (right) for all five runs. Deep C reaches the lowest loss and highest accuracy; the B settings trail their A counterparts.

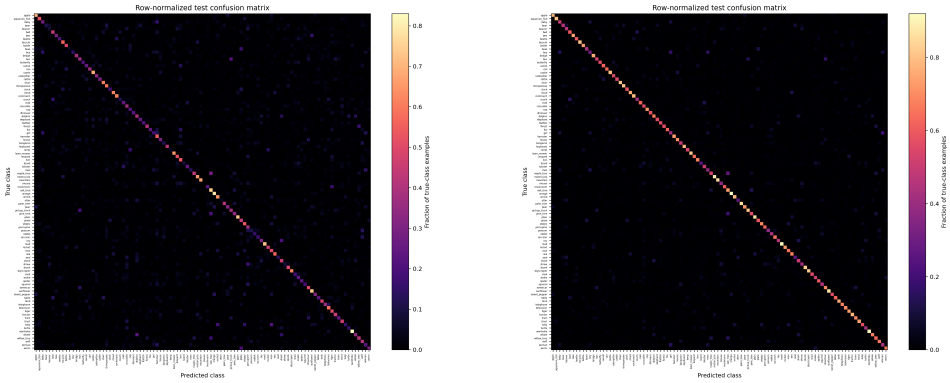


Fig. 2. Row-normalized test confusion matrices for Compact A (left) and Deep C (right). Rows are true classes, columns are predictions, and the diagonal is class recall.

attributions are summed over RGB and each map is normalized by its own maximum, so brightness is not comparable across images.

Examples come from the selected model (Deep C) using a fixed rule: scan classes for one with at least two correct predictions and one error, then take its highest-confidence correct, lowest-confidence correct, and highest-confidence incorrect examples. The rule selects true class *apple* (test indices 7331, 2803, and 2093).

High-confidence correct (99.95%). Attribution is concentrated on the upper and central part of the fruit, including around its bright highlight, with little on the background. This is consistent with an object-focused decision, though a map alone does not prove a semantic "apple" concept.

Low-confidence correct (34.57%). The apple is correct but attribution is sparse and spread over the fruit and the pale surroundings, consistent with a less decisive representation; the map does not show that any one factor caused the lower confidence.

High-confidence error (97.75%, sweet pepper). The image contains two red apples and one yellow apple, and attribution concentrates on the yellow one rather than the background. The model therefore attends to an object region yet maps it to the wrong class, and color or shape resemblance may contribute, but IG cannot establish that mechanism.

IG is descriptive rather than causal here: absolute values

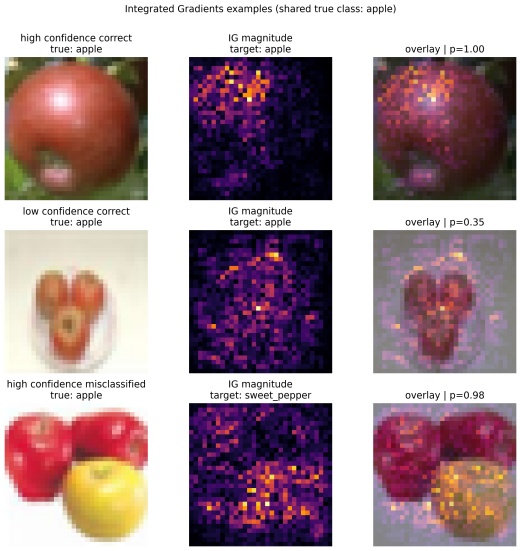


Fig. 3. IG for Deep C on three true-*apple* test images: high-confidence correct (99.95%), low-confidence correct (34.57%), and high-confidence error predicted as *sweet pepper* (97.75%). Columns: input, IG magnitude, overlay.

discard sign, per-image scaling blocks magnitude comparison, a black baseline is not neutral for every image, softmax saturation may affect confident predictions, and no alternative baselines or completeness checks were run.

VI. Strengths and Limitations

Strengths are a reproducible balanced split, two model capacities across multiple settings, selection by validation loss only, and complementary evaluation (accuracy, top-5, Wilson intervals, curves, confusion matrices, IG). Limitations are a single random seed (intervals do not capture retraining variance), jointly varied hyperparameters, Setting C tested only on Deep, only two architecture families, and IG on three examples. Future work should vary learning rate, dropout, and L2 independently, repeat the best settings over several seeds, isolate depth from width, and extend IG with signed attributions and alternative baselines.

VII. Conclusion

Deep C is the best model, with 63.04% top-1 accuracy (95% CI 62.09–63.98%), 36.96% generalization error (CI 36.02–37.91%), and 88.66% top-5 accuracy. Deep outperforms Compact by about 26 points across settings, lighter regularization (C vs. A) adds only a statistically unclear 0.29 points, and the lower-learning-rate, heavier-regularization Setting B reduces accuracy for both architectures. Confusion matrices confirm broader class-level gains for Deep, and IG shows that relevant object regions can drive correct and incorrect predictions alike, including a confident apple-to-sweet-pepper error.

References

[1] A. Krizhevsky, "Learning multiple layers of features from tiny images," University of Toronto, Tech. Rep., 2009. [Online]. Available: https://www.cs.toronto.edu/~kriz/cifar.html

[2] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," in *Proc. 34th International Conference on Machine Learning (ICML)*, vol. 70, 2017, pp. 3319–3328.